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Research Article

ASENIDSB: A Stacked Ensemble-based Network Intrusion Detection System Optimized with Boruta Feature Selection

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Abstract

In the cyber world, network security is increasingly challenged by sophisticated attacks and vulnerabilities that exploit data and systems. Effective security solutions are crucial for real-time protection, yet existing methods often struggle with imbalanced datasets, redundant features, and limited classification accuracy. Network Intrusion Detection Systems (NIDS) play a vital role in combating these threats, but traditional ML approaches frequently fall short in detecting complex and subtle attack patterns. This research addresses these critical issues by introducing a Stacked Ensemble-Based Network Intrusion Detection System using Boruta (ASENIDSB). This model tackles the problem of class imbalance, a significant challenge in IDS, by employing the K-means SMOTE oversampling technique to improve the representation of minority classes. Additionally, the Boruta feature selection method improves computational efficiency by addressing redundancy and inefficiency brought on by irrelevant features. ASENIDSB also improves classification robustness and detection capabilities using a stacked ensemble learning technique. It integrates Extra Trees and Extreme Gradient Boosting as base classifiers, with Logistic Regression serving as the meta-model to deliver precise and reliable predictions. This combination enables the model to address the limitations of single and traditional ensemble classifiers, offering a robust, efficient, and accurate solution for network intrusion detection. The accuracy and F1-score of the proposed ASENIDSB model are evaluated using a ten-fold cross-validation method. Experimental Results have proved that the proposed model significantly surpasses some other ensemble models, standard ML classifiers, and SOTA models. The proposed ASENIDSB model achieved an impressive accuracy of 97.85% and an F1-score of 97.82%.

Keywords

Network Intrusion Detection System, stacked ensemble, boruta, K-means SMOTE, isolation Forest, robust Scaler

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1 Introduction

Unauthorized access or harmful behavior on a computer network or system by an outside attacker or an insider is referred to as a network intrusion. Network intrusions typically involve attempts to gain unauthorized access to system or network resources, steal confidential data, disrupt operations, or install malware or backdoors for future attacks. Common examples of network intrusions include denial-of-service (DoS) attacks, brute-force attempts, malware infections, port scanning, phishing, botnet attacks, and social engineering techniques. Although these examples illustrate the broad landscape of network threats, this research focuses specifically on analyzing and mitigating intrusion behaviors based on the categories present in the employed dataset. Businesses can suffer significantly from network intrusions, leading to monetary losses, reputational damage, legal consequences, and the loss of sensitive information (Caporale et al., 2023).

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IDS that are specifically made to look at autonomous system activity, including keeping an eye on log files, are known as host-based IDS. When implemented, HIDS will examine all incoming and outgoing packets for the system. It also keeps an eye on the host system's operating system (OS). The file system's previous image will be compared with a picture taken by the HIDS of the full file system set. If the comparison results show that the file system has changed or been modified, the administrator will be notified. The Sagan, OSSEC, and other tools are used for HIDS (Hossain & Islam, 2023). Examining network traffic from every host connected to the network is the purpose of the network-based intrusion detection system. The traffic is analyzed over the entire subnet and contrasted with the traffic those assaults have already passed through. The network administrator will receive a warning or alert if the comparison finds any dangers. The primary NIDS tools include Snort, Bro, and other similar ones.

Network Intrusion Detection Systems (NIDS) are essential components of modern cybersecurity frameworks, designed to monitor network activity and detect malicious behaviors such as unauthorized access or data breaches. NIDS provide automated solutions to identify and respond to network attacks and security vulnerabilities, making them critical for ensuring the confidentiality, integrity, and availability of sensitive data in networked environments (Rajadurai & Gandhi, 2022). Despite their importance, traditional Intrusion Detection Systems (IDS) face significant challenges, including high false positive rates, difficulty in detecting previously unseen attack patterns, and scalability issues in handling large and complex datasets (Sandosh et al., 2020).

The integration of Machine Learning (ML) into IDS has shown promise in addressing these challenges. ML-based models enable intelligent, adaptive intrusion detection by leveraging data-driven insights to identify and classify threats. Single-classifier models such as Support Vector Machines (SVM), Decision Trees (DT), and K-Nearest Neighbors (KNN) have been extensively used. However, these models are prone to overfitting, suffer from high variance or bias, and often struggle with imbalanced datasets, where certain attack classes are underrepresented (Sarkar et al., 2023). To overcome these limitations, ensemble learning methods, which combine multiple classifiers to improve performance, have gained popularity. Techniques such as Extra Trees (ET), Gradient Boosting (GB), and Extreme Gradient Boosting (XGB) are widely employed for intrusion detection, offering enhanced accuracy and robustness. However, ensemble models still face challenges, including sensitivity to outliers and the need for optimal feature selection to improve computational efficiency and accuracy (Bukhari et al., 2023).

In recent years, addressing class imbalance, optimizing feature selection, and enhancing model accuracy through advanced ensemble methods have become critical challenges in developing effective network intrusion detection systems. This research focuses on integrating these strategies into a comprehensive solution to significantly improve detection performance. This study presents an innovative model ASENIDSB (A Stacked Ensemble-Based Network Intrusion Detection System using Boruta). ASENIDSB's principal contributions consist of:

1. Class Imbalance Handling and Outlier Detection: To improve detection performance, the model incorporates K-means SMOTE for addressing class imbalance and Isolation Forest for effective outlier detection, ensuring reliable data representation and enhanced classification accuracy.
2. Feature Optimization: The Boruta evolutionary feature selection technique is employed to reduce computational cost and maintain high predictive performance by selecting the most relevant features, thereby improving model efficiency.
3. Advanced Ensemble Learning Architecture: A stacked ensemble learning framework is developed, combining Extreme Gradient Boosting (XGB) and Extra Trees (ET) as base learners with Logistic Regression (LR) as the meta-classifier, to deliver accurate and dependable intrusion detection outcomes.

By solving current shortcomings in class imbalance, anomaly management, and computing efficiency, these advances establish ASENIDSB as a reliable and effective network intrusion detection solution. Ten-fold cross-validation is used to validate the proposed ASENIDSB model, and essential metrics like accuracy and F1-score are used to assess its performance. Comparative analysis with single-classifier models, other ensemble-based models, and state-of-the-art methods highlights the superiority of ASENIDSB in terms of accuracy, robustness, and efficiency. This study provides a scalable and practical solution for detecting and mitigating network intrusions, contributing to advancements in cybersecurity research and practice.

The paper's following sections are arranged: Section 2 examines pertinent methods already studied. The proposed methodology is thoroughly explored in Section 3. Section 4 discusses and presents the findings and outcomes of the proposed method. The paper is finally concluded in section 5. The future scope is directed in section 6.

2 Literature Review

As the demand for an effective network analytical solution has grown, several researchers have helped to improve intrusion detection system designs and establish several machine learning-based hybrid models for security attack detection. Rules that consistently scan for threats and compare them to a list of known threats were the basis of the first model made public for the IDS (Mishra & Paliwal, 2023). Al-Zubaidie and Shyaa (2023) proposed a hybrid cryptographic model aimed at enhancing transaction security in e-commerce applications by detecting cryptographic leakage. The work addresses the limitations of traditional cryptographic systems, which are often susceptible to side-channel attacks and data leakage, particularly in mobile and online transaction environments. Although their approach does not rely on conventional machine learning classifiers, it incorporates anomaly detection and leakage analysis techniques to identify potential vulnerabilities and enhance data confidentiality. The detailed Literature Review with limitations is described in the table 1.

2.1 Review Summary

The reviewed literature highlights the growing reliance on machine learning techniques to address challenges in network intrusion detection systems (NIDS), particularly in handling imbalanced datasets, feature selection, and real-time threat detection. Various studies propose ensemble-based methods, such as stacked ensembles, tree-based models, and multi-layer frameworks, to improve detection accuracy and computational efficiency. Techniques like SMOTE and hybrid feature selection methods, including genetic algorithms and recursive elimination, are widely used to mitigate class imbalance and optimize features. While these approaches demonstrate improved performance, limitations persist in terms of scalability, adaptability to dynamic threats, and computational overhead in high-dimensional or large-scale datasets. Advanced solutions such as bio-inspired algorithms, multi-dimensional feature fusion, and anomaly detection using outlier-based methods provide promising directions but often require further refinement to address evolving cybersecurity needs effectively. Overall, the literature underscores the importance of integrating robust feature selection, class balancing, and ensemble techniques to enhance the practicality and reliability of NIDS in diverse network environments.

2.2 Bibliometric Analysis

The proportionate distribution of reviewed publications from major academic databases, including IEEE, Springer, Elsevier, and other repositories, is shown in Figure 1(a). The distribution of reviewed articles by category, divided into conference proceedings, book chapters, and journals, is shown in Figure 1(b). The percentage distribution of papers across several topic areas, such as machine learning, ensemble learning, and other pertinent approaches, is shown in Figure 1(c). Using carefully chosen keywords ('Intrusion detection OR Network Intrusion detection OR Intrusion detection system') AND ('Machine Learning OR Ensemble Learning'), Figure 1(a)–(c) collectively shows the curated dataset of "Scholarly Works" aggregated from prestigious academic repositories such as Scopus, IEEE Xplore, and other digital libraries. Finding useful insights that can help new and emerging scholars in this field is the major goal of this data aggregation.

3 The Proposed Methodology

The proposed ASENIDSB model consists primarily of six key steps: i. Dataset preparation, ii. Threat Model iii. Handling Class Imbalance by utilizing K- Means SMOTE, iv. Outlier Detection using Isolation Forest, v. Feature selection using Boruta, vi. Normalization using Robust Scaler, and vii. Classification using XGB and ET stacked ensemble method.

Figure 2 illustrates the flow diagram of the proposed ASENIDSB model.

3.1 Dataset Description

The University of Nevada Reno - Intrusion Detection Dataset (UNR-IDD) 2023[1] is used in this study for multiclass classification to address various network-based threats. The data set consists of six classes, including a 'normal' class, and contains 32 features (Das et al., 2023). To avoid biased results, duplicate entries have been removed. The features are categorized into 3 main groups: Port Statistics, from Feature-1 to Feature-9; Delta Port Statistics, i.e., Feature-10 to Feature-18; and Flow Entry and Flow Table Statistics, which refers to features Feature-19 through Feature-32.

Feature Categories in the UNR-IDD 2023 Dataset

Port Statistics (F1–F9): Includes features like received packets, sent bytes, and packet errors.

Delta Port Statistics (F10–F18): Tracks changes in port statistics over time, such as delta received bytes and delta packets dropped.

Flow Entry and Flow Table Statistics (F19–F32): Covers features like connection point, active flow entries, and packets matched.

Table 1. Articles and sources of Intrusion Detection System.

Author	Research Objectives	Limitations of Existing System	Machine Learning Techniques Used
Chowdhury et al. (2023)	Develop a bi-phase NIDS using real-time traffic analysis.	Focused on real-time traffic but lacks emphasis on scalability and handling imbalanced datasets.	Random Forest, Decision Tree
Kharwar and Thakor (2022)	Propose an ensemble approach combining feature selection and classification using Extra Tree algorithm.	Limited evaluation on datasets with diverse and modern attack patterns.	Extra Trees, Ensemble Feature Selection
Rashid et al. (2022)	Employ a tree-based stacking ensemble technique with feature selection for NIDS.	High computational cost and lacks consideration for dynamic attack evolution.	Stacking Ensemble, Feature Selection (Boruta)
Zhang et al. (2021)	Propose multi-dimensional feature fusion with stacking ensemble for NIDS.	High-dimensional data handling needs further optimization.	Stacking Ensemble, Feature Fusion
Shafieian and Zulkernine (2023)	Develop multi-layer stacking ensemble learners for low-resource NIDS.	Limited application to high-throughput network environments.	Stacking Ensemble
Lian et al. (2020)	Use decision tree-based recursive feature elimination in ensemble learning for intrusion detection.	Scalability and dynamic attack adaptability are not addressed.	Decision Tree, Recursive Feature Elimination
Gu et al. (2019)	Implement SVM ensemble with feature augmentation for intrusion detection.	High computational complexity with imbalanced datasets.	SVM Ensemble
Moustafa et al. (2018)	Propose ensemble techniques using statistical flow features for IoT network traffic protection.	Focused on IoT networks; lacks broader applicability to traditional network environments.	Statistical Flow Features, Ensemble Methods
Rajagopal et al. (2020)	Design a stacking ensemble for NIDS using heterogeneous datasets.	Limited scalability for real-time intrusion detection.	Stacking Ensemble
Wu et al. (2022)	Combine enhanced random forest with SMOTE for intrusion detection.	Computational cost increases with SMOTE; limited exploration of feature selection.	Random Forest, SMOTE
Turukmane and Devendiran (2024)	Propose M-MultiSVM with efficient feature selection for NIDS.	Computationally expensive and limited evaluation on imbalanced data.	Multi-SVM
Venkatesan (2023)	Design an IDS focusing on feature selection using ML algorithms.	It requires better handling of modern attack complexities and imbalanced data.	Feature Selection, Various ML Algorithms
Bakro et al. (2024)	Build a cloud-based IDS using hybrid bio-inspired feature selection and Random Forest.	Focuses on cloud environments but not traditional network setups.	Random Forest, Bio-Inspired Feature Selection
Kavitha et al. (2023)	Apply filter-based ensemble feature selection with deep learning for cloud computing intrusion detection.	High resource requirements; limited comparison with other deep learning methods.	Deep Learning, Ensemble Feature Selection
Talukder et al. (2024)	Use oversampling, stacking feature embedding, and feature extraction for NIDS on big data.	Limited applicability to resource-constrained environments.	Stacking Ensemble, SMOTE
Widodo et al. (2024)	Apply SMOTE for multi-class imbalanced datasets in intrusion detection.	Lack of feature selection optimization.	SMOTE, Ensemble Learning
Kudithipudi et al. (2023)	Evaluate resampling techniques for addressing class imbalance in NIDS using SVM.	Resampling increases computational overhead and has limited scalability to large datasets.	SVM, Resampling Techniques
ELhadad et al. (2023)	Compare Enhanced Isolation Forest and Local Outlier Factors for anomaly detection.	Limited application to NIDS datasets and real-world scenarios.	Isolation Forest, Local Outlier Factor

(Continued)

Table 1. Continued.

Author	Research Objectives	Limitations of Existing System	Machine Learning Techniques Used
Esteban (2023)	Combine Ant Colony Optimization with deep learning for anomaly recognition in wireless ad-hoc networks.	Focused on ad-hoc networks; lacks generalizability to other network setups.	Ant Colony Optimization, Deep Learning
Das et al. (2021)	Perform a comparative analysis of ensemble ML techniques with feature selection for NIDS.	High resource consumption; requires further analysis of modern attack patterns.	Ensemble Methods, Feature Selection
Sayem et al. (2024)	Propose ENIDS, a deep learning-based ensemble framework for NIDS.	Computationally expensive with limited real-time testing.	Deep Learning, Ensemble Framework

The dataset's features are outlined in the table below:

Attack Types in the Dataset

The dataset includes five types of attack alongside normal traffic, representing various threats originating from network devices and hosts:

1. **TCP-SYN Flood:** This attack exploits the TCP three-way handshake by sending a high volume of SYN packets without completing the handshake, overwhelming the server's resources and leading to service unavailability. TCP-SYN Flood attacks create significant challenges for intrusion detection systems due to their high traffic similarity to legitimate SYN requests.
2. **Port Scan:** Port scanning attempts to identify open ports and vulnerable services on a target system. While often part of the reconnaissance phase of a cyberattack, distinguishing between benign and malicious scanning requires careful analysis of traffic patterns and connection attempts.
3. **Flow Table Overflow:** By generating numerous unique flows, attackers can overflow the flow tables of network switches, disrupting normal forwarding processes. This type of attack is particularly critical in Software Defined Networks (SDN), where control plane resources are limited.
4. **Blackhole Attack:** In a blackhole attack, network nodes are compromised to silently drop data packets, preventing them from reaching their intended destinations. Detection is difficult because dropped packets can also occur due to legitimate network issues.
5. **Traffic Diversion:** This attack manipulates routing information to redirect traffic, often enabling interception or delay of data delivery. It poses a serious threat in environments where maintaining data integrity and confidentiality is critical.

This dataset is utilized to train machine learning classifiers for multi-class intrusion detection, addressing various network-based threats.

3.2 Threat Model

To clearly define the adversarial context and security assumptions of the proposed ASENIDSB framework, a threat model has been developed based on the attack types represented in the UNR-IDD 2023 dataset. This threat model helps contextualize the goals and capabilities of potential attackers and aligns the system design with realistic network-layer threats.

– **Adversary Objectives:** The attacker aims to compromise network integrity or availability by injecting malicious traffic or manipulating routing behavior, targeting vulnerabilities in SDN-based architectures.
– **Attack Vectors:** The UNR-IDD 2023 dataset includes five primary attack types that reflect the threat landscape:
  - **TCP-SYN Flood:** Launches volumetric DoS attacks by overwhelming the server with incomplete TCP handshakes.
  - **Port Scan:** Probes host systems to identify open ports for potential exploitation.
  - **Flow Table Overflow:** Exploits limited flow table capacity in SDN switches by flooding with fake entries.
  - **Blackhole Attack:** Silently drops traffic without forwarding, disrupting normal communication.
  - **Traffic Diversion:** Redirects data flows to unauthorized nodes for interception or modification.

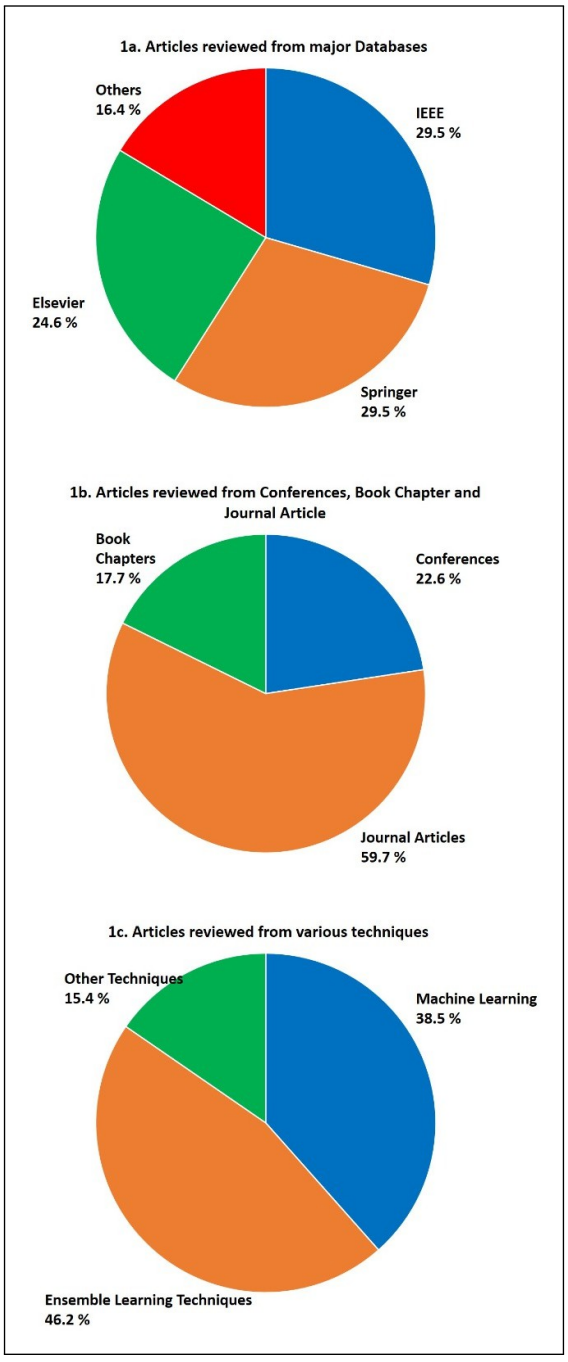


Figure 1. (a) Articles Reviewed from Major Databases (b) Articles Reviewed from Conferences, Book Chapters, and Journal Articles (c) Articles Reviewed from Various Techniques.

- **Attacker Capabilities:** The attacker can inject or alter traffic at the network level but cannot access the internal logic or training phase of the intrusion detection model.
- **System Assumptions:** The network is monitored by an NIDS deployed at a strategic point in the SDN infrastructure. It analyzes flows extracted via OpenFlow for real-time attack detection.
- **Security Objective:** The ASENIDSB model is designed to detect and classify these attack scenarios with high accuracy while maintaining robustness against class imbalance and irrelevant features.

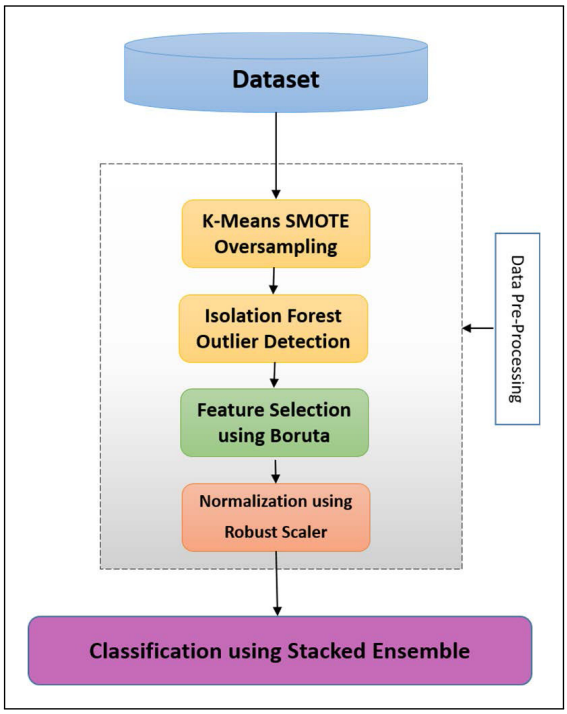


Figure 2. Flowchart for the ASENIDSB Methodology.

3.3 Class Imbalance Treatment Using K- Means SMOTE Oversampling

The UNR-IDD dataset exhibits a significant class imbalance, where normal samples greatly outnumber abnormal ones. This skew disrupts the training of detection models, leading to strong performance on the majority class but poor accuracy for minority classes, ultimately weakening the model's generalization ability. For example, rare attack types like TCP-SYN flood or flow table overflow are underrepresented, making detection less reliable.

To address this challenge, K-means SMOTE is utilized. Traditional oversampling techniques like SMOTE often amplify the influence of outliers, which can distort the data distribution and reduce model performance. K-means SMOTE first applies K-means clustering to group similar data points and isolate outliers. It ensures that synthetic samples are generated from more representative data clusters. After clustering, the minority samples are used to calculate the mean of neighboring points, which is the basis for creating synthetic data points. This method minimizes the influence of outliers while enriching the diversity of the new samples.

Compared to other approaches like Borderline-SMOTE or SMOTE-ENN, K-means SMOTE offers several advantages. It reduces overfitting by maintaining better control over sample distribution, preserves the inherent characteristics of the data, and generates synthetic samples that align more closely with real-world attack patterns. It improves detection performance for minority classes and enhances the classification model's overall robustness and generalization capability. As a result, implementing K-means SMOTE ensures a more balanced dataset, enabling the model to address better the challenges posed by the diverse attack scenarios in the UNR-IDD dataset.

3.4 Outlier Detection Using Isolation Forest

Outliers, or data points that significantly differ from normal patterns, can negatively impact the performance of machine learning models, especially in fields like intrusion detection and fraud detection. In the UNRIDD dataset, which contains network traffic data for intrusion detection, outliers often arise due to variations in attack types (e.g., PortScan, TCP-SYN) and normal traffic patterns. These outliers, if not effectively managed, can lead to inaccurate model predictions, higher false positives, and overfitting.

To address this issue, Isolation Forest (iForest) is applied. Unlike traditional methods that model "normal" data and detect deviations, Isolation Forest isolates anomalies by constructing decision trees (Isolation Trees, ITs) that partition data randomly. Anomalous points are identified based on their shorter path lengths in the tree structure, making them

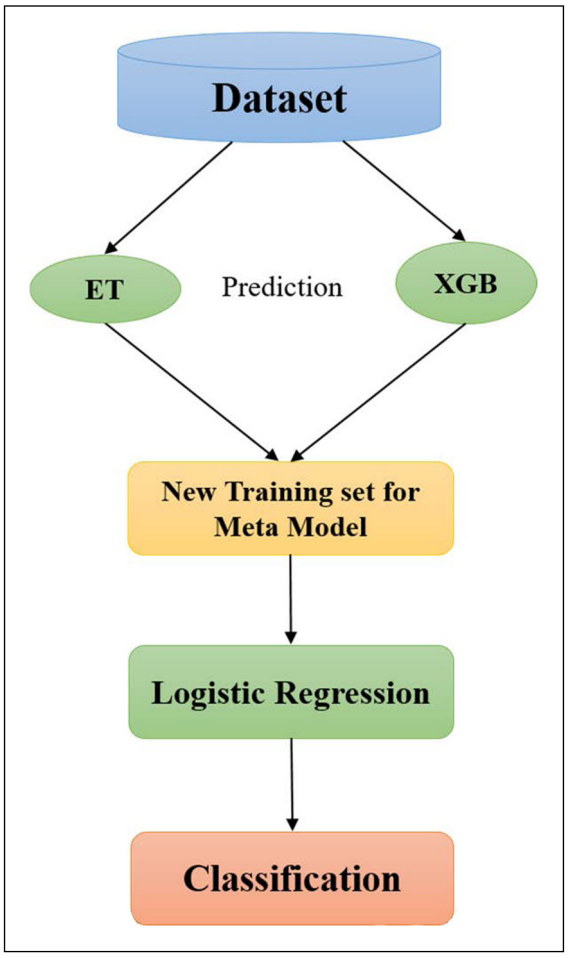


Figure 3. The General Outline of the Proposed Method.

easier to isolate. This method is particularly effective for high-dimensional, large-scale datasets like UNRIDD due to its ability to detect outliers without assumptions about data distribution.

Isolation Forest offers several advantages over other anomaly detection techniques. It is computationally efficient, scalable to large datasets, and does not rely on distributional assumptions, making it ideal for complex network traffic data. Unlike methods such as k-means or Gaussian Mixture Models, which may struggle with noisy or high-dimensional data, Isolation Forest effectively isolates anomalies, making it a suitable solution for improving the performance of intrusion detection systems.

3.5 Feature Selection Using Boruta

High dimensionality in network traffic data poses a major challenge for machine learning algorithms, as it not only hampers scalability but also restricts their ability to generalize effectively. This challenge is especially pronounced in the UNRIDD dataset, which contains a large number of features, many of which are irrelevant or redundant. The sheer volume of features in this dataset leads to sparsity in the feature space, making it difficult to analyze and interpret data patterns accurately. As a result, the model's performance suffers from reduced classification accuracy, higher computational costs, and slower convergence during training. One effective strategy to address this problem is feature selection, which reduces the dimensionality of the dataset by identifying and retaining the most relevant features. In particular, the Boruta algorithm has proven to be highly effective for this purpose, as it helps in identifying important features while eliminating irrelevant ones. By leveraging randomization and selecting multiple samples in an ensemble-based manner, Boruta minimizes the impact of random fluctuations and correlations, providing a clearer view of the significant features. This not only enhances classification accuracy but also reduces the computational complexity of feature extraction, making it an ideal solution for handling the challenges associated with high-dimensional data in the UNRIDD dataset. Additionally, feature selection

technique, such as Boruta, is known to optimize model performance by improving accuracy, reducing overfitting, and lowering computational demands, as highlighted in previous studies.

In this study, the Boruta feature selection technique is implemented using Python on the UNRIDD dataset, which initially contained 32 features. By leveraging the Boruta library, 17 relevant features are identified, and 15 irrelevant attributes are discarded. The Boruta algorithm, based on random forest classifiers, is used to assess the significance of each feature, ensuring that only the most impactful features are retained.

The Boruta algorithm selects features by comparing their importance to that of randomly shuffled copies, called shadow features. It expands the dataset with these, calculates feature importance using a Random Forest, and determines the Maximum Z-score of shadow Attributes (MZSA). This MZSA score acts as a dynamic threshold: Real features with significantly higher Z-scores are deemed "important," while those significantly lower are "unimportant" and removed. Features with unclear results undergo further evaluation. This iterative process ensures that only features with demonstrably higher predictive power than random noise are retained, resulting in a more reliable and efficient feature set for the model.

The Boruta algorithm consists of the following steps:

1. Make sure the information system is continuously extended with at least five shadow attributes by expanding the database by duplicating all variables, even if the initial set contained fewer than five characteristics.
2. To eliminate any associations with the result, rearrange the remaining properties.
3. Run a random forest classifier using the extended information system, then note the calculated Z scores.
4. Assign each feature a "hit" that exceeds the MZSA score after determining which shadow property has the most excellent Z score (MZSA).
5. Conduct a two-sided equality test for each attribute with an undetermined degree of relevance using the MZSA score.
6. If a feature's significance dramatically falls short of the MZSA score, mark it as "unimportant" and permanently remove it from the information system.
7. If an attribute's relevance significantly surpasses the MZSA score, classify it as "important."
8. Take into account none of the shadow qualities.
9. Keep going until the algorithm has used up all of the random forest runs or all the characteristics have been ranked, whichever comes first.

3.6 Normalization Using Robust Scaler

Normalization is a crucial preprocessing step in machine learning, as datasets often contain features with varying scales, which can bias algorithms toward larger values. Without normalization, models may perform poorly or converge slowly. Z-score normalization and Min-Max scaling are common techniques. Min-Max scaling rescales the data between 0 and 1 using the formula:

$$x_{\text{scaler}} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{1}$$

Data are adjusted to have a mean of 0 and a standard deviation of 1 using Z-score normalization, also known as standardization. The Robust Scaler improves on these methods by normalizing based on the interquartile range (IQR), making it less sensitive to outliers. The formula is as follows.

$$x = \frac{x_i - Q_1(x)}{Q_3(x) - Q_1(x)} \tag{2}$$

where Q_1 (x) and Q_3 (x) are the 25th and 75th percentiles. For datasets like UNR-IDD, which contain outliers and varying feature scales, the Robust Scaler is ideal choice as it reduces the influence of extreme values, ensuring stable and accurate model performance for intrusion detection.

3.7 Classification Using XGB and ET Stacked Ensemble Method

Stacking, also known as a multiple classifier system, is an advanced ensemble learning technique that combines predictions from multiple base classifiers to improve accuracy and generalization. In this approach, diverse base classifiers (e.g., B1, B2,. . . , Bn) are trained on the dataset to generate outputs, which are then aggregated into a new dataset for a second-level classifier, called the meta-learner. The meta-learner refines these predictions by identifying and correcting errors made by the base classifiers, enabling more accurate classification of unseen data.

Stacking can utilize homogeneous or heterogeneous base learners, offering flexibility in model selection. The process involves k-fold cross-validation to train and validate the meta-learner, reducing overfitting and enhancing performance. Unlike traditional methods where a single classifier may struggle with bias or variance, stacking leverages the strengths of multiple models, making it highly effective for handling complex datasets.

Algorithm 1. Stacked Ensemble Learning Algorithm

Require: Training dataset $D = \{(a_1, b_1), (a_2, b_2), \ldots, (a_n, b_n)\}$, Feature set $F = \{f_1, f_2, \ldots, f_n\}$, Number of trees $nTrees = 100$

Ensure: Predicted class labels (attacked or normal)

Step 1: Train the first base learner (XGB)

Initialize $f_0(a)$ using:

$$f_0(a) = \arg\min \sum_{i=1}^{m} (\gamma - b_i)^2$$

for $i = 1$ to m **do**

Compute the residual error:

$$g_i(a) = \text{predicted value} - \text{actual value}$$

Update the model:

$$f_i(a) = f_{i-1}(a) + g_i(a)$$

end for

Step 2: Train the second base learner (Extra Trees)

Initialize an empty set G

for $i = 1$ to $nTrees$ **do**

Select a subset $T(i)$ of F

Identify the most important feature g_i from $T(i)$

Update G:

$$G \leftarrow G \cup \{g_i\}$$

end for

Step 3: Create a new dataset using predictions from base learners

for $i = 1$ to n **do**

Generate predictions $h_1(a), h_2(a), \ldots, h_n(a)$ from base learners

Store $M_h = \{(a_1, b_1), (a_2, b_2), \ldots, (a_n, b_n)\}$

end for

Step 4: Train the meta-classifier (Logistic Regression)

Use M_h to train the final classifier

Step 5: Perform classification

Predict the class label (attacked or normal) for each instance using the meta-classifier

Standard machine learning classifiers often face challenges such as high bias, variance, and class imbalance, which limit their effectiveness in real-world applications. High bias leads to underfitting, while high variance results in overfitting. Class imbalance, a common issue in datasets like UNRIDD, causes models to favor the majority class, neglecting critical minority class instances. Ensemble models like Extra Trees (ET), Random Forest (RF), Gradient Boosting (GB), and XGB address these challenges by combining multiple classifiers, improving accuracy and robustness. However, they have limitations; for instance, boosting techniques such as XGB are prone to overfitting noisy data, while bagging methods such as ET may struggle with imbalanced datasets.

This work introduces ASENIDSB (A Stacked Ensemble-Based Network Intrusion Detection System using Boruta) to overcome these issues. ASENIDSB incorporates advanced preprocessing techniques: Isolation Forest for outlier detection, K-Means SMOTE for addressing class imbalance, and Boruta for selecting the most relevant features. These steps ensure a high-quality and balanced data set for classification.

The stacking ensemble architecture of ASENIDSB combines ET and XGB as base classifiers and Logistic Regression (LR) as the meta-classifier. ET handles variance effectively with its randomized splits, while XGB captures complex

patterns through gradient boosting. LR, as the meta-classifier, aggregates predictions from the base models into a well-calibrated probabilistic output. This setup leverages the strengths of each classifier to improve generalization. By employing k-fold cross-validation, ASENIDSB minimizes overfitting and systematically corrects errors from base classifiers. The result is a robust model that effectively handles outliers, class imbalance, and irrelevant features, achieving superior classification accuracy and reliability in real-world network intrusion detection tasks.

4 Experimental Results and Analysis

The Experimental Results and Evaluation in the proposed paper are structured into five sections. The performance metrics are described in Section A. Section B focuses on evaluating the performance of the proposed model, ASENIDSB, and its comparative analysis with other standard ML classifiers. Section C provides a comparison of the ASENIDSB model with alternative ensemble models. In Section D, the proposed ASENIDSB model is compared with other models and techniques reported in the literature. Finally, Section E discusses the effectiveness of the ASENIDSB model in detail. A comprehensive explanation of each section is provided below:

4.1 Performance Metrics

Accuracy is deemed the most significant performance metric in the proposed approach. The GBM-RF model's performance in an intrusion detection system is evaluated by this accuracy. The accuracy of classification (AC), detection rate (DR), precision, and recall are among the measures that have been used to assess the effectiveness of the proposed method. Eqs. 3 through 6 represent these measures as true positive (TP), true negative (TN), false positive (FP), and false negative (FN), depending on the order (Rajadurai & Gandhi, 2022).

$$\text{Accuracy} = \sum_{i=1}^{n} \frac{(TP_i + TN_i)}{(TP_i + TN_i + FP_i + FN_i)} \tag{3}$$

$$Precision_i = \sum_{i=1}^{n} \frac{TP_i}{TP_i + FP_i} \tag{4}$$

$$\text{Recall}_i = \sum_{i=1}^{n} \frac{TP_i}{TP_i + FN_i} \tag{5}$$

$$F1\ \text{score} = \sum_{i=1}^{n} \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{6}$$

4.2 Performance of ASENIDSB and Comparison with Various Standard ML Classifiers

From the dataset containing 32 independent features, the Boruta feature selection technique was applied to identify the most significant features for the classification task. This method effectively selected 17 critical features while discarding the remaining 15, which are deemed irrelevant or redundant. Using these refined features, the stacked ensemble model is trained to evaluate the performance of the proposed model, ASENIDSB (A Stacked Ensemble-Based Network Intrusion Detection System using Boruta).

The evaluation of the ASENIDSB model revealed remarkable results, achieving an F1-score of 97.82% and an accuracy of 97.85%, demonstrating its robustness and reliability in classifying network intrusions. These performance metrics are illustrated in Table 3 and Figure 4 for better visualization and understanding. To further highlight the significance of the proposed model's performance, ASENIDSB is compared against several standard ML classifiers. These included Linear Discriminant Analysis (LDA), Decision Trees (DT), Naïve Bayes (NB), Logistic Regression (LR), and K-nearest neighbor (KNN). Each comparison incorporated identical data preprocessing steps to ensure fairness and consistency in evaluation. This comparative analysis underscores the efficiency and superiority of the ASENIDSB model in handling network intrusion detection tasks.

As shown in Table 3 and Figure 4, the proposed model, ASENIDSB, outperforms standard ML classifiers in terms of both F1-score and accuracy. This improvement stems from the limitations of standard ML classifiers, which struggle to address the variance and bias inherent in the data. In contrast, ASENIDSB effectively manages both variance and bias by leveraging ET to combine predictions from multiple DT models and incorporating the SMOTE oversampling technique.

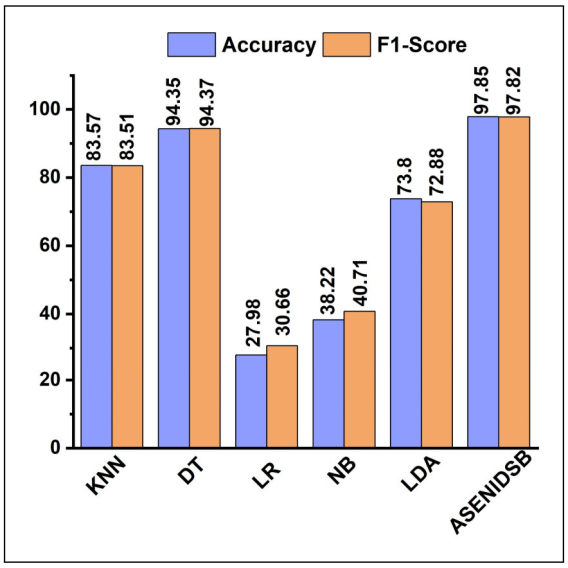


Figure 4. Bar Graph Depicting F1-Score and Accuracy of ASENIDSB Compared to Various Standard ML Classifiers on the UNR-IDD Dataset.

Table 2. The UNR-IDD Dataset's Features.

No.	Feature	No.	Feature
1	Packets Received	17	Delta Packets Rx_Errors
2	Bytes Received	18	Delta Packets Tx_Errors
3	Packets Sent	19	Connection Point
4	Bytes Sent	20	Total Load/Rate
5	Port_alive Duration	21	Total Load/Latest
6	Packets Rx_Dropped	22	Unknown Load/Rate
7	Packets Tx_Dropped	23	Unknown Load/Latest
8	Packets Rx_Errors	24	Time_seen
9	Packets Tx_Errors	25	Is valid
10	Received Delta Packets	26	Table ID
11	Received Delta Bytes	27	Active Flow Entries
12	Delta Sent Packets	28	Packets Looked Up
13	Delta Sent Bytes	29	Packets Matched
14	Delta Port Alive Duration	30	Max Size
15	Delta Packets Rx_Dropped	31	Time_Seen
16	Delta Packets Tx_Dropped	32	Is valid

4.3 Comparison of ASENIDSB's Performance with Different Ensemble Models

After completing all the data preprocessing stages, the proposed ASENIDSB model, is evaluated against ensemble models such as Extreme Gradient Boosting (XGB), Random Forest (RF), CatBoost, and Gradient Boosting (GB). The results of the comparison are presented in Table 4 and Figure 5 below.

As observed in Table 4 and Figure 5, the proposed model, ASENIDSB, outperforms the ensemble models in terms of F1-score and accuracy. This superior performance is attributed to the use of ET in ASENIDSB, which effectively manages dataset variance. In contrast, boosting methods like GB and XGB struggle to address variance adequately. Although bagging ensembles like RF can handle variance, they underperform compared to ET. This is because ET minimizes variance more effectively by employing randomization to select the best split at each node, resulting in faster computations. Moreover, Extra Trees maintains robust performance even in the presence of noisy features and enhances classification accuracy by aggregating the outputs of multiple de-correlated decision trees within a "forest."

Table 3. Performance Comparison of ASENIDSB and Standard ML Classifiers on the UNR-IDD Dataset.

Model	Accuracy	F1-Score
KNN	83.57	83.51
DT	94.35	94.37
LR	27.98	30.66
NB	38.22	40.71
LDA	73.80	72.88
ASENIDSB	**97.85**	**97.82**

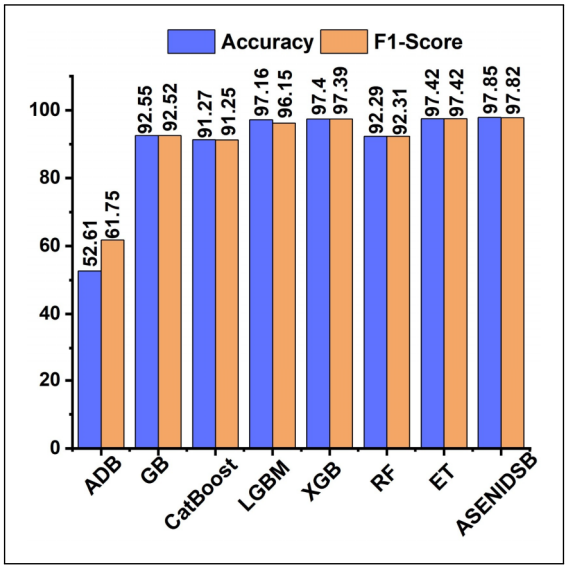


Figure 5. ASENIDSB and other Ensemble Models' Accuracy and F1-score Bar Graphs Utilizing the UNR-IDD Dataset.

Table 4. F1-score and Accuracy Comparison of other Ensemble Models with the Proposed Model.

Model	Accuracy	F1-Score
ADB	52.61	61.75
GB	92.55	92.52
CatBoost	91.27	91.25
LGBM	97.16	96.15
XGB	97.40	97.39
RF	92.29	92.31
ET	97.42	97.42
ASENIDSB	**97.85**	**97.82**

4.4 Comparison of ASENIDSB's Performance with SOTA Models

The proposed ASENIDSB model is compared with models Abbas et al. (2022), Kharwar and Thakor (2022), Bhati and Rai (2020), Sharma and Yadav (2021) and the results are shown below in Table 5 and Figure 6

Comparing the performance of the proposed ASENIDSB model with other state-of-the-art models such as Abbas et al. (2022), Kharwar and Thakor (2022), Bhati and Rai (2020), and Sharma and Yadav (2021), Table 5 and Figure 6 show that the proposed approach performs remarkably better than the previously stated models. The models above did not address the issue of class imbalance; for feature selection consistency (CONS), they employ a feature selection technique that is somewhat arbitrary and slower than Boruta. Additionally, for classification, they employ Hidden Naïve Bayes. The accuracy of this standard ML model is substantially worse than that of the proposed ASENIDSB model since it cannot handle the bias and variance in the dataset.

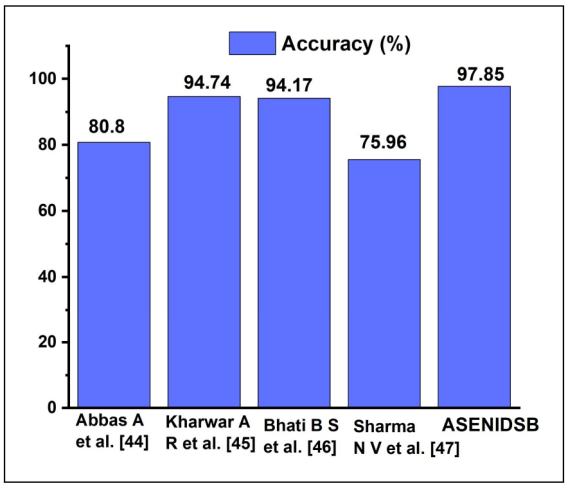


Figure 6. Accuracy and F1-Score Bar Graph for ASENIDSB and Other SOTA Models.

Table 5. Performance of ASENIDSB Compared with Various SOTA Models.

Model	Accuracy
Abbas et al. (2022)	80.80
Kharwar and Thakor (2022)	94.74
Bhati and Rai (2020)	94.17
Sharma and Yadav (2021)	75.96
ASENIDSB	**97.85**

Table 6. Comparison of ASENIDSB's Performance with that of ASENIDSB without Feature Selection and ASENIDSB without Oversampling.

Model	Accuracy	F1-Score
ASENIDSB without pre-processing	96.50	96.49
ASENIDSB without oversampling	97.08	97.10
ASENIDSB without feature selection	97.36	97.33
ASENIDSB	**97.85**	**97.82**

4.5 Ablation Study of the Proposed ASENIDSB Model

When the proposed ASENIDSB model is compared to its counterpart without feature selection, the results indicate a notable enhancement in performance. This improvement is attributed to eliminating feature redundancy, which reduces noise and improves the model's ability to focus on the most relevant attributes for classification. Similarly, a comparison between ASENIDSB and its variant without the oversampling technique reveals a significant reduction in class imbalance, leading to improved model performance. By addressing the issue of minority class underrepresentation, oversampling ensures a more balanced training dataset, which enhances the accuracy and reliability of the model's predictions. The detailed comparative results are presented in Table 6 and illustrated in Figure 7 below.

4.6 Evaluation on Additional Datasets

The proposed ASENIDSB model is evaluated with three other datasets, such as KDD Cup 99, NSL-KDD, and CICIDS2018, to assess its generalizability and robustness. The results of each of these datasets are shown Table 7 and Figure 8. The results confirm the model's consistent performance across diverse network intrusion detection datasets, demonstrating strong adaptability to varying data distributions.

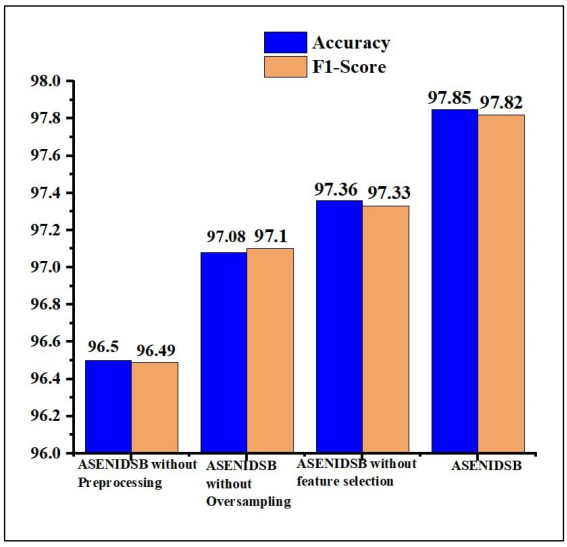


Figure 7. A Bar Graph Showing How Well ASENIDSB Performs in Comparison to ASENIDSB without Oversampling and ASENIDSB without Feature Selection.

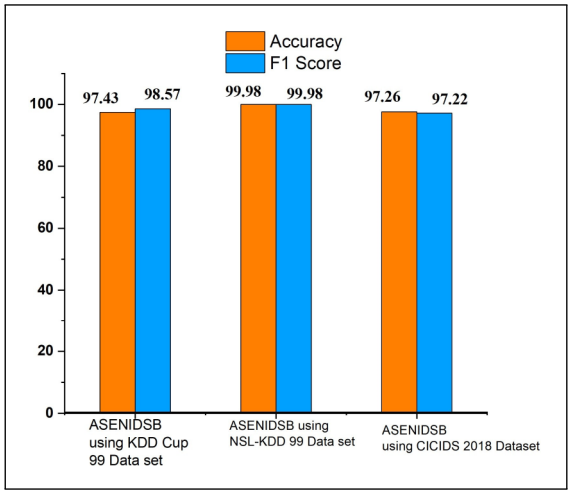


Figure 8. A Bar Graph Showing How Well ASENIDSB Performs in Comparison to other Datasets such as KDD Cup 99, NSL-KDD, and CICIDS2018.

4.7 Discussion & Analysis

While the stacked ensemble nature of ASENIDSB offers superior predictive performance, the interpretability of its decision-making process can be a challenge inherent in complex models. However, our approach incorporates elements that provide valuable insights for network administrators. The Boruta feature selection method yields a clear ranking of feature importance, enabling administrators to identify the key network attributes indicative of malicious activity. This information can guide monitoring efforts, inform security policy adjustments, and aid in understanding the characteristics of different attack vectors. Furthermore, the actionable output of the ASENIDSB model, classification of network activity as normal or malicious with an associated confidence score, allows for timely responses and prioritization of security alerts. Future work could explore the application of Explainable AI (XAI) techniques to provide more granular insights into the model's predictions and further enhance its utility for network administrators in understanding and mitigating security threats.

As shown in Figure 9, the bar graph presents the top 17 features identified by the Boruta algorithm as the most influential in the classification task. Notably, features such as 'Delta Port Alive Duration (S)', 'Total Load/Latest', and 'Unknown

Table 7. Comparison of Proposed ASENIDSB Model's Performance with other Datasets Such as KDD Cup 99, NSL-KDD, and CICIDS2018.

Model	Accuracy	F1-Score
ASENIDSB using KDD Cup 99 Dataset	97.43	98.57
ASENIDSB using NSL-KDD Dataset	99.98	99.98
ASENIDSB using CICIDS 2018 Dataset	97.26	97.22

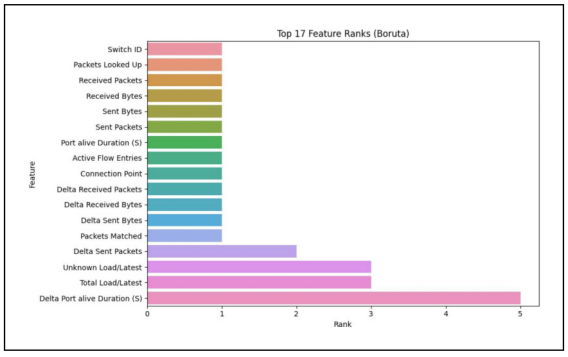


Figure 9. Feature Importance Rankings Generated by the Boruta Feature Selection Algorithm for the ASENIDSB Model.

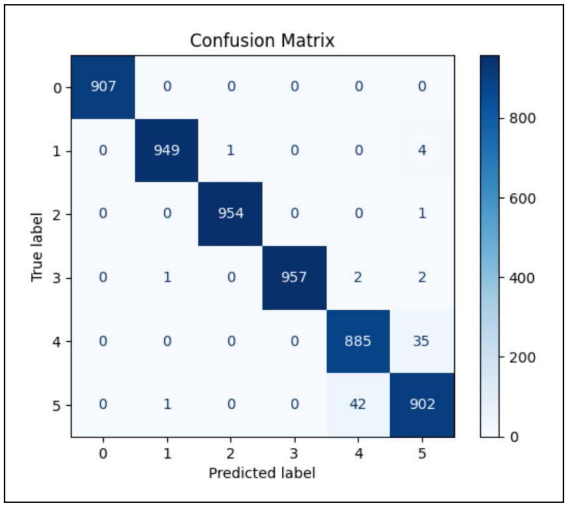


Figure 10. Confusion Matrix of the Proposed ASENIDSB Model Illustrating the Classification Results Across Different Attack Categories.

Load/Latest' are identified as highly influential. This ranking aids in eliminating redundant features and optimizing the proposed ASENIDSB model performance by focusing on the most relevant attributes.

In Figure 10, the confusion matrix produced by the ASENIDSB model on the UNR-IDD 2023 dataset shows how well the six classes such as Normal, TCP-SYN Flood, Port Scan, Flow Table Overflow, Blackhole Attack, and Traffic Diversion performed in classification. For each class, the diagonal components show instances that have been accurately classified, and the off-diagonal elements show instances that have been misclassified. The model performs quite well on the normal and major attack categories, and it exhibits good accuracy across all classes. Although Blackhole and Traffic Diversion attacks may have similar traffic patterns, there are some minor misclassifications between them. Overall, the confusion matrix confirms the robustness and effectiveness of ASENIDSB in multiclass intrusion detection.

The multi-class ROC-AUC curves for the ASENIDSB model evaluated using the UNR-IDD 2023 dataset are shown in Figure 11. The model's ability to differentiate one class from the others is shown by each curve. Excellent discriminating power is indicated by the consistently high Area Under the Curve (AUC) values for all six classes, which range from 0.96

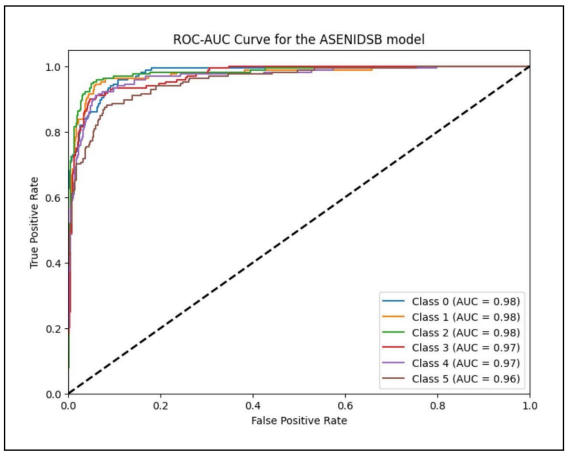


Figure 11. ROC-AUC Curve of the Proposed ASENIDSB Model for Multiple Attack Classes.

to 0.98. The ASENIDSB model successfully reduces false positives while preserving a high true positive rate across all attack categories and the normal class, as evidenced by the curves' proximity to the top-left corner.

4.7.1 Limitations. While the ASENIDSB model demonstrates strong performance across several metrics, certain limitations remain. One potential concern is overfitting, especially due to the complexity introduced by ensemble learning and oversampling techniques like K-means SMOTE. Although cross-validation was employed to mitigate this, overfitting risk persists in highly dynamic or unseen network environments. Scalability is another consideration; the stacked ensemble structure, along with computational overhead from Boruta feature selection and K-means clustering, may pose challenges for real-time deployment on large-scale or high-throughput networks. Additionally, the model's performance is sensitive to hyperparameter tuning–suboptimal configurations can significantly affect detection accuracy and generalization. Therefore, future work should focus on lightweight optimization methods and adaptive learning strategies to address these constraints effectively.

4.7.2 Real-World Deployment Considerations. While the proposed ASENIDSB model demonstrates strong performance in experimental settings, practical deployment in real-world environments presents several challenges. One key consideration is computational complexity, particularly due to ensemble learning and the use of multiple preprocessing techniques such as K-means SMOTE, and Boruta feature selection. These processes, while effective, may increase training and inference times, potentially affecting latency in real-time detection scenarios. Furthermore, resource-constrained environments such as edge devices or low-power systems may face difficulties in hosting the full model pipeline without optimization. Integration with existing network security infrastructure is another practical aspect, requiring compatibility with real-time traffic monitoring systems, log management tools, and alerting mechanisms. To mitigate these challenges, future work will focus on model compression, pruning, and the use of lightweight ensemble variants. Additionally, seamless deployment can be supported through the development of APIs or containerized microservices, ensuring scalability and ease of integration with current security frameworks.

4.7.3 Strengths of ASENIDSB Compared to Deep Learning Approaches. Interpretability: The Boruta feature selection technique, which ASENIDSB uses, offers a transparent feature priority ranking. This contrasts with deep learning models, which are frequently referred to as "black boxes" because of their intricate internal representations, which make it challenging to determine which elements have the greatest predictive power.

Computational Efficiency: The ASENIDSB model is likely to be more computationally efficient than deep learning models, which typically require significant computational resources for training, especially when dealing with large datasets.

Less Data Requirement: The proposed ASENIDSB model can perform well with smaller datasets as well as large datasets, whereas deep learning models often require large amounts of data to achieve high performance.

Robustness to Overfitting: The use of ensemble techniques and cross-validation in ASENIDSB can provide robustness against overfitting, a common issue in deep learning models that may require additional regularization techniques to address.

4.7.4 Weaknesses of ASENIDSB Compared to Deep Learning Approaches. Scalability to Very Large Datasets: Deep learning models, particularly those based on neural networks, are known for their scalability and performance with very large datasets. ASENIDSB, while efficient, may not scale as effectively as deep learning models when dealing with extremely large and complex datasets.

Feature learning: Deep learning models are able to automatically learn complex feature representations from raw data. ASENIDSB relies on manual feature selection and may not capture as complex feature interactions as deep learning models, which can automatically learn and combine features in a hierarchical fashion.

Real-time data processing: Some deep learning models, especially those designed for real-time analytics, can process streaming data efficiently. ASENIDSB may require batch processing and might not be as effective for real-time intrusion detection without further optimization.

Adaptability: Deep learning models can be more adaptable to new and unseen types of attacks, especially if they are designed with online learning capabilities. ASENIDSB might require retraining or updating the ensemble components to maintain its effectiveness against new threats.

5 Conclusion

This study introduces the ASENIDSB intrusion detection model, which was developed and evaluated on the UNR-IDD dataset and initially converted the categorical features into numerical features to train the data using the pandas library. The K-means SMOTE oversampling technique addresses the class imbalance. The isolation forest outlier detection technique is applied to remove the outliers from the balanced dataset. The Boruta feature selection technique is used to identify relevant features, enhancing model performance by reducing overfitting and lowering computational demands. The proposed ASENIDS model then processes the refined data with 17 selected features. ASENIDS employs base classifiers such as XGB, Extra Trees, which overcomes variance, and the LR Meta classifier. A comparative analysis evaluated the proposed model's accuracy and F1-score on the UNR-IDD dataset, leveraging 10-fold cross-validation to measure the model's effectiveness. Results are then benchmarked against those obtained from other ensemble classifiers, some SOTA models, and standard ML classifiers. As illustrated in Tables 2–4 and Figures 4–6, the proposed ASENIDSB model significantly outperformed these models, achieving an impressive accuracy of 97.85% and an F1-score of 97.82%. These results underscore ASENIDSB's potential as an effective tool for robust network intrusion detection. Moreover, the proposed model demonstrates improved differentiation between benign and specific attack types in a multiclass classification context.

6 Future Scope

Future research can expand ASENIDSB by testing it on more extensive and diverse data sets to ensure scalability and adaptability to real-world scenarios. Real-time implementation and dynamic adaptation to evolving threats can enhance its practical effectiveness. Further improvements in multiclass classification and integration with emerging AI techniques, like graph-based networks or transformers, may bolster its accuracy and robustness. Additionally, lightweight and energy-efficient versions could enable deployment in resource-constrained environments such as IoT. Lastly, exploring domain-specific customizations and resilience against adversarial attacks could enhance its utility across various industries and critical infrastructure. By enhancing privacy, energy efficiency, and adversarial robustness, federated learning will significantly increase its performance and applicability to changing cybersecurity threats.

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Note

1. https://github.com/anandhumohan/UNR-IDD-Classification

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